Experiment - 1

Title : Design an ER Diagram for an Indian E-Commerce Platform.

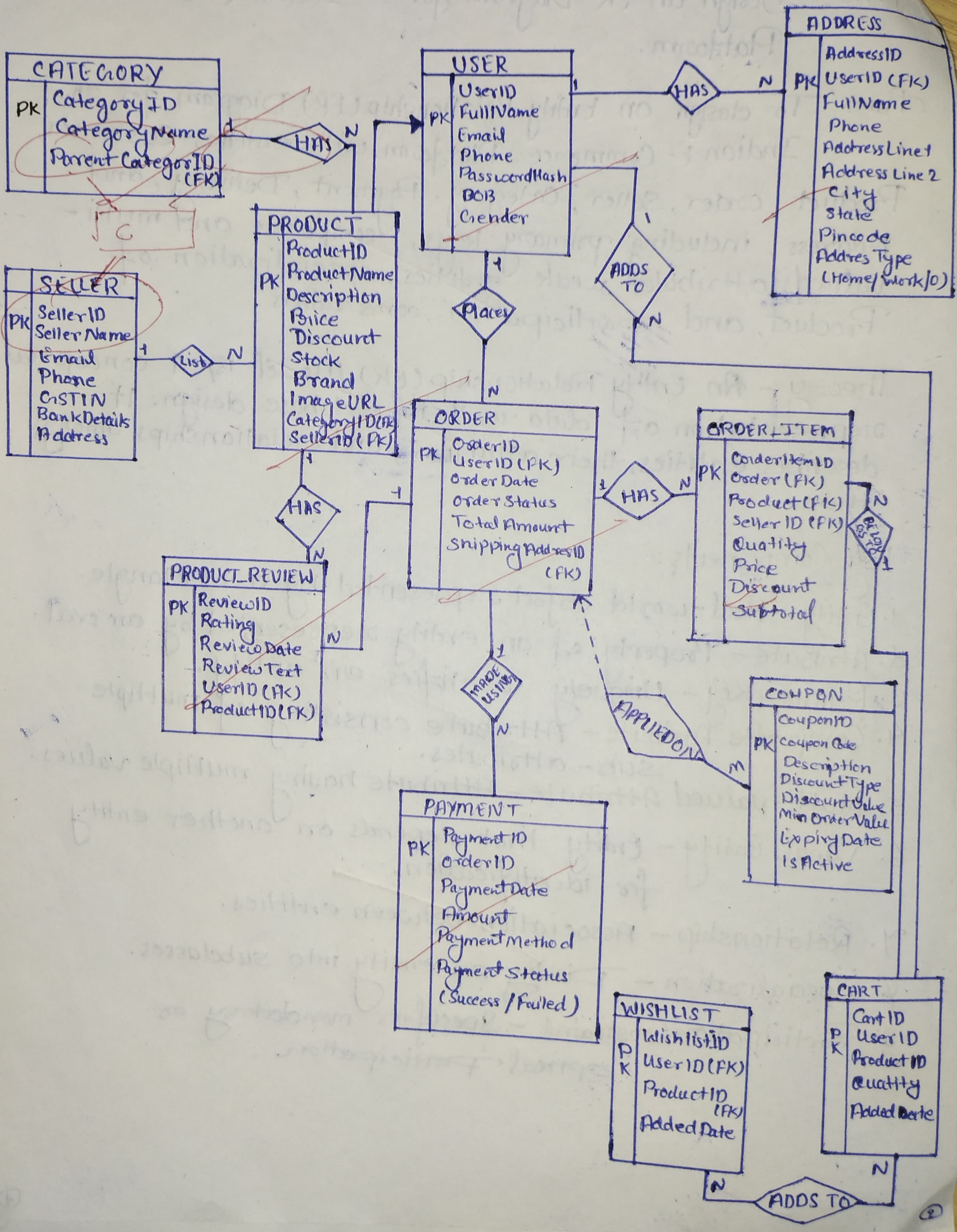
Aim:- To design an Entity Relationship (ER) Diagram for an Indian E-Commerce Platform with entities Customer, Product, Order, Seller, Category, Payment, Delivery, and Address, including primary keys, composite and multi-valued attributes, weak entities, specialization of Product, and participation constraints.

Theory:- An Entity Relationship (ER) Model is a conceptual representation of data used in database design. It describes entities, their attributes, and relationships among them.

Key Components:-

1. Entity - Real-world object represented by a rectangle.
2. Attribute - Property of an entity represented by an oval.
3. Primary Key - Uniquely identifies an entity.
4. Composite Attribute - Attribute consisting of multiple sub-attributes.
5. Multi-valued Attribute - Attribute having multiple values.
6. Weak Entity - Entity that depends on another entity for identification.
7. Relationship - Association between entities.
8. Specialization - Dividing an entity into subclasses.
9. Participation Constraint - Specifies mandatory or optional participation.

ER-DIAGRAM


CATEGORY
PK
CategoryID
CategoryName
ParentCategoryID (FK)
HAS
1
N
USER
PK
UserID
FullName
Email
Phone
PasswordHash
DOB
Gender
HAS
1
N
ADDRESS
PK
AddressID
UserID (FK)
FullName
Phone
AddressLine1
AddressLine2
City
State
Pincode
AddressType (Home/Work/Other)
ADDS TO
1
N
PRODUCT
PK
ProductID
ProductName
Description
Price
Discount
Stock
Brand
ImageURL
CategoryID (FK)
SellerID (FK)
SELLER
PK
SellerID
SellerName
Email
Phone
GSTIN
BankDetails
Address
LIST
1
N
PLACES
1
N
HAS
1
N
ORDER
PK
OrderID
UserID (FK)
OrderDate
OrderStatus
TotalAmount
ShippingAddressID (FK)
HAS
1
N
ORDER_ITEM
PK
OrderItemID
Order (FK)
Product (FK)
SellerID (FK)
Quantity
Price
Discount
Subtotal
REFERS TO
N
1
PRODUCT_REVIEW
PK
ReviewID
Rating
ReviewDate
ReviewText
UserID (FK)
ProductID (FK)
1
N
MADE USING
1
N
APPLIED ON
N
M
COUPON
PK
CouponID
CouponCode
Description
DiscountType
DiscountValue
MinOrderValue
ExpiryDate
IsActive
PAYMENT
PK
PaymentID
OrderID
PaymentDate
Amount
PaymentMethod
PaymentStatus
(Success / Failed)
WISHLIST
PK
WishlistID
UserID (FK)
ProductID (FK)
AddedDate
CART
PK
CartID
UserID
ProductID
Quantity
AddedDate
N
N
ADDS TO